Definition 1.9

Let $f : X \to Y$ be a bijection. The function $g : Y \to X$ defined by $g(y) = x$ if $f(x) = y$, is called the ***inverse*** of f and is denoted by f^{-1}.

If a function f has an inverse, then we say that f is ***invertible***. There is a nice relationship between composition of functions and inverse.

Let $f : X \to Y$ be a bijection and $g : Y \to X$ be its inverse. Then $g \circ f = I_X$ and $f \circ g = I_Y$ where I_X and I_Y are identity functions on X and Y respectively. Moreover, if $f : X \to Y$ and $g : Y \to X$ are functions such that $g \circ f = I_X$ and $f \circ g = I_Y$, then both f and g are bijections and they are inverses to each other; that is $f^{-1} = g$ and $g^{-1} = f$.

Using the discussions above, the terms invertible and inverse can be defined in some other way as follows:

Definition 1.10

A function $f : X \to Y$ is said to be ***invertible*** if there exists a function $g : Y \to X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$ where I_X and I_Y are identity functions on X and Y respectively. In this case, g is called the inverse of f and g is denoted by f^{-1}.

We may use this concept to prove some functions are bijective.

If f is a bijection, then $f^{-1}(y)$ is nothing but the pre-image of y under f. Let us note that the inverses are defined only for bijections. If f is not one-to-one , then there exists a and b such that $a \neq b$ and $f(a) = f(b)$. Let this value be y. Then we cannot define $f^{-1}(y)$ because both a and b are pre-images of y under f, as f^{-1} cannot assume two different values for y. If f is not onto, then there will be a y in Y without a pre-image. In this case also we cannot assign any value to $f^{-1}(y)$.

For example, if $A = \{1, 2, 3, 4\}$ and $f = \{(1, 2), (2, 4), (3, 1), (4, 3)\}$. Then the range of f is $\{1, 2, 3, 4\}$; the inverse of f is $\{(1, 3), (2, 1), (3, 4), (4, 2)\}$.

Working Rule to Find the Inverse of Functions from $\mathbb{R}$ to $\mathbb{R}$:

Let $f : \mathbb{R} \to \mathbb{R}$ be the given function.

i. write $y = f(x)$;
ii. write x in terms of y;
iii. write $f^{-1}(y) =$ the expression in y.
iv. replace y as x.

Example 1.30 If $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = 2x - 3$ prove that f is a bijection and find its inverse.

Solution:
Method 1:

One-to-one : Let $f(x) = f(y)$. Then $2x-3 = 2y-3$; this implies that $x = y$. That is, $f(x) = f(y)$ implies that $x = y$. Thus f is one-to-one.

Onto : Let $y \in \mathbb{R}$. Let $x = \frac{y+3}{2}$. Then $f(x) = 2(\frac{y+3}{2}) - 3 = y$. Thus f is onto. This also can be proved by saying the following statement. The range of f is $\mathbb{R}$ (how?) which is equal to the co-domain and hence f is onto.

Inverse Let $y = 2x - 3$. Then $y + 3 = 2x$ and hence $x = \dfrac{y+3}{2}$. Thus $f^{-1}(y) = \dfrac{y+3}{2}$. By replacing y as x, we get $f^{-1}(x) = \dfrac{x+3}{2}$.